

INDIAN INSTITUTE OF TECHNOLOGY, KHARAGPUR
End-Autumn Semester 2025-26

Date of Exam: 11/25
Subject Number: MA60225
Subject Name: Financial Mathematics

Session(FN/AN)
Max. Marks: 50

Duration: 3 hrs
Department: Mathematics
No. of students: 126

Instructions:

- (i) Use of calculator and Statistical tables is allowed.
- (ii) All answers of numerical questions must be in **at least two decimal places**.
- (iii) All the notations are standard. If any data/statement is missing, identify it on your answer script.
- (iv) Answer **All** questions.
- (v) All parts of a question **Must Be** answered at **One Place**.

1. For this question (for question 1 only) write only answers on the first page of your answer script in the given tabular form. Detail working may be carried out on other pages. [1×5]

Q. No. 1	a	b	c	d	e
Answer					

Let $\{W_t, t \geq 0\}$ be a Wiener process.

- (a) Compute the variance of stochastic integral $\int_0^1 t^4 dW_t$.
- (b) Apply Itô's formula to compute the stochastic differential $d(e^t W_t)$.
- (c) Is the process $Y_t := e^t W_t, t \geq 0$, a Martingale? (Yes/No)
- (d) Let $T = \text{Min}\{t : W_t = 2 - 3t\}$, that is T is the first time the Wiener process hits the line $2 - 3t$. Use Martingale stopping theorem to calculate $E(T)$.
- (e) Consider a European call option on a stock that is currently selling for Rs. 80. The option expires in 1 year at a strike price of Rs. 85. Suppose that the risk-free rate is 5% and that the volatility is $\sigma = 0.15$ per year. Compute the value of the call.

2. (a) Give the interpretation of the Greeks (delta, gamma, theta) of European call in Black-Scholes framework.

(b) Derive put call parity.

(c) Argue that Black-Scholes market is arbitrage-free.

~~(d)~~ A straddle is a portfolio consisting of a call and a put on the same underlying stock, with the same strike and same expiry. Plot the graph of the payoff function (of the terminal stock price) of the straddle (on one share of the stock).

~~(e)~~ Explain value at risk (VaR).

[5×3]

3. ~~(a)~~ Using binomial model derive Black-Scholes formula. Then explain Black-Scholes framework.

~~(b)~~ Consider a two period binomial financial market with $u = 5/3$, $d = 1/3$ and $r = 0$. Let the current (time $t = 0$) stock price be $S_0 = 60$. Price a put with maturity $T = 2$ and strike $K = 80$.

[5+5]

~~4~~ Consider a $T = 2$ period binomial financial market with $u = 1.35$, $d = 0.6$ and $r = 0.15$. Let the current (time $t = 0$) stock price be $S_0 = 400$,

- ~~(i)~~ Price a call option with maturity $T = 2$ and strike $K = 500$.
- ~~(ii)~~ Construct a replicating portfolio for this option.
- ~~(iii)~~ directly verify that the replicating strategy you found in part (ii) is self-financing.
- ~~(iv)~~ price a put with the same maturity and strike price.

[10]

~~5~~ Derive the sensitivities "delta, gamma, theta, vega and rho" of European call in Black-Scholes framework.

[10]
